Notes: Restuccia and Rogerson (RED, 2008)

Policy Distortions and Aggregate Productivity with Heterogeneous Establishments

Contents

1 Big picture

- **Question.** How much can aggregate output and measured total factor productivity fall when policies distort the prices faced by individual producers and thereby misallocate resources across heterogeneous establishments?
- **Setting.** The paper embeds heterogeneous establishments into a neoclassical growth model with entry, exit, and decreasing returns at the plant level, and then calibrates the undistorted benchmark to U.S. data.
- **Core challenge.** Standard growth accounting typically focuses on aggregate capital accumulation and aggregate relative prices. This paper argues that even if aggregate prices and aggregate factor stocks are unchanged, distortions that vary across producers can still create large efficiency losses by moving labor and capital away from their efficient plant-level allocation.
- **Main theoretical result.** Idiosyncratic taxes or subsidies distort establishment size. With decreasing returns to scale, this creates a wedge between the decentralized allocation and the efficient one, lowering measured TFP.
- **Main quantitative result.** In the benchmark calibration, distortions that are negatively correlated with productivity can lower output and measured TFP by roughly 30% to 50%, even when aggregate capital accumulation is held fixed.
- **Key comparative-static insight.** "Noise" distortions that are unrelated to plant productivity have smaller effects unless they apply to a very large fraction of establishments. Distortions that systematically subsidize low-productivity plants or tax high-productivity plants are much more damaging.
- **Why the paper matters.** It became a benchmark for the modern misallocation literature because it shows, in a transparent calibrated framework, that cross-country income differences need not come only from low aggregate capital or low frontier technology. They can also come from poor allocation across producers.

2 Data and measurement (key objects)

2.1 Calibration targets from U.S. data

This is a calibrated theory paper rather than a reduced-form micro-econometric paper. The "data" enter by disciplining the undistorted benchmark economy.

- **Time period.** One model period is one year.
- **Benchmark country.** The United States is treated as the undistorted economy.
- **Targets from aggregate data.** The model matches a real interest rate of 4%, an investment-output ratio of 20%, and standard capital and labor income shares.
- **Targets from establishment data.** The calibration uses the U.S. distribution of establishment employment sizes, especially the fact that most establishments are very small while a small set of large establishments account for a disproportionately large share of output and employment.

- **Size range.** Employment at the establishment level is taken to range from 1 to 10,000 workers, which disciplines the support of establishment productivity.

2.2 Model-level objects observed or constructed

- **Establishment productivity.** Each plant draws a time-invariant productivity level s.
- **Policy distortion.** Each plant faces a tax or subsidy τ that changes the effective price of output or, in extensions, the effective price of capital or labor.
- **Invariant distribution.** The steady-state object of interest is the mass of establishments by type, $\mu(s, \tau)$, together with aggregate entry E.
- **Aggregate outcomes.** Output, capital, wages, entry, and measured TFP are all computed in steady state under alternative policy regimes.
- **Reallocation summary statistics.** The paper tracks the output share of subsidized establishments Y_s/Y, the subsidy bill as a fraction of output S/Y, and the subsidy rate τ_s required to satisfy the experiment's constraint.

2.3 Empirical applications inside the paper

The paper uses the calibrated model for four main quantitative exercises:

- uncorrelated idiosyncratic distortions,
- correlated idiosyncratic distortions,
- correlated distortions when aggregate capital is allowed to change,
- and alternative distortions operating through capital rental rates or wages rather than output prices.

3 Models and empirical specifications (all equations + interpretation)

3.1 Household side and feasibility

There is an infinitely lived representative household with utility

$$\sum_{t=0}^{\infty} \beta^t u(C_t),$$

where $0 < \beta < 1$.

Aggregate feasibility is

$$C_t + X_t + c_e E_t \leq Y_t - M_t c_f,$$

where C_t is consumption, X_t is investment, E_t is entry, M_t is the mass of producing establishments, and c_f is the fixed operating cost per active establishment.

Capital evolves according to

$$K_{t+1} = (1-\delta)K_t + X_t.$$

- **Interpretation.** This is a standard growth-model backbone, but aggregate production is not generated by one representative plant. Instead it is the sum of heterogeneous establishments.
- **Why this matters.** Once plants differ in productivity, aggregate output depends not just on total capital and labor but also on *who gets them*.

3.2 Plant technology, entry, and exit

Each establishment has production function

$$Y = f(s,k,n) = sk^{\alpha}n^{\gamma}, \qquad \alpha,\gamma \in (0,1), \qquad \alpha+\gamma < 1.$$

- s is establishment-specific productivity.
- k is capital services and n is labor services.
- Decreasing returns to scale, $\alpha+\gamma<1$, are crucial because they pin down an optimal plant size.

Plants also face:

- a fixed operating cost c_f,
- an exogenous death probability λ,
- and a one-time entry cost c_e.

New establishments draw productivity from a discrete distribution

$$s \in \{s_1,\dots,s_{n_s}\}, \qquad h(s) = \Pr(\text{draw } s).$$

- **Interpretation.** The model is essentially a Hopenhayn-style environment with entry, exit, and plant heterogeneity, embedded in a growth setting.
- **Important simplification.** The only difference across plants is TFP level s. In the undistorted benchmark, all plants therefore use the same capital-labor ratio.

3.3 Policy distortions

The core distortion is an establishment-specific output tax or subsidy τ. The plant receives effective after-tax revenue $(1-\tau)\times$ output. Negative values of τ represent subsidies.

A policy rule is summarized by

$$P(s,\tau),$$

which gives the probability that a plant with productivity s faces distortion τ. Combined with the productivity distribution $h(s)$, this induces the joint distribution

$$g(s,\tau) = h(s)P(s,\tau).$$

- **Interpretation.** The policy can be unrelated to productivity, or systematically correlated with it.
- **Key economic distinction.** An uncorrelated distortion creates "noise" in plant-level prices. A correlated distortion actively moves resources toward the wrong productivity types.
- **Government budget.** The fiscal consequences of taxes and subsidies are closed through lump-sum transfers or taxes to the representative household, so the analysis isolates allocation effects rather than labor-supply effects.

3.4 Steady-state equilibrium

The household Euler equation implies the steady-state rental rate of capital:

$$r = \frac{1}{\beta} - (1 - \delta).$$

Define the corresponding real interest rate

$$R = r - \delta = \frac{1}{\beta} - 1.$$

- **Interpretation.** The consumer side pins down the aggregate rental rate; the rest of the equilibrium problem is then about the distribution of activity across establishments.

3.5 Incumbent plant optimization

Given prices (w, r) and distortion τ, an incumbent plant solves

$$\pi(s, \tau) = \max_{n,k \geq 0} \Big\{ (1 - \tau) s k^{\alpha} n^{\gamma} - wn - rk - c_f \Big\}.$$

The optimal factor demands are

$$\bar{k}(s, \tau) = \left(\frac{\alpha}{r}\right)^{\frac{1-\gamma}{1-\alpha-\gamma}} \left(\frac{\gamma}{w}\right)^{\frac{\gamma}{1-\alpha-\gamma}} \big(s(1 - \tau)\big)^{\frac{1}{1-\alpha-\gamma}},$$

$$\bar{n}(s, \tau) = \left(\frac{\gamma}{w}\right)^{\frac{1-\alpha}{1-\alpha-\gamma}} \left(\frac{\alpha}{r}\right)^{\frac{\alpha}{1-\alpha-\gamma}} \big(s(1 - \tau)\big)^{\frac{1}{1-\alpha-\gamma}}.$$

The value of an incumbent is

$$W(s, \tau) = \frac{\pi(s, \tau)}{1 - \rho}, \qquad \rho = \frac{1 - \lambda}{1 + R}.$$

- **Interpretation.** Output taxes or subsidies work like plant-specific TFP shifters, because they change the effective profitability of operating a plant of type s.
- **Key implication.** When two plants have the same physical productivity but different τ, they choose different scales. With decreasing returns, that is inefficient.

3.6 Entry, invariant distribution, and market clearing

A potential entrant pays c_e, draws (s, τ), and then decides whether to operate. The value of entry is

$$W^e = \sum_{(s,\tau)} \max\{W(s,\tau), 0\} g(s,\tau) - c_e.$$

With free entry,

$$W^e = 0.$$

If $\bar{x}(s,\tau) \in \{0,1\}$ indicates whether a type-(s,τ) entrant operates, the invariant distribution of establishments is

$$\mu(s,\tau) = E\,\frac{\bar{x}(s,\tau)}{\lambda} g(s,\tau).$$

Steady-state market clearing requires

$$1 = \sum_{(s,\tau)} \bar{n}(s,\tau)\mu(s,\tau),$$

$$K = \sum_{(s,\tau)} \bar{k}(s,\tau)\mu(s,\tau),$$

$$C + \delta K + c_e E = \sum_{(s,\tau)} \big(f(s,\bar{k},\bar{n}) - c_f\big)\mu(s,\tau),$$

$$T + \sum_{(s,\tau)} \tau f(s,\bar{k},\bar{n})\mu(s,\tau) = 0.$$

- **Interpretation.** Given r, free entry pins down the wage w, and labor-market clearing pins down the mass of entry E.
- **Why entry matters.** Even though the main focus is reallocation across active establishments, distortions can also change how many plants operate, and therefore affect average scale and aggregate outcomes through entry.

3.7 Calibration logic

The calibration treats the U.S. as the distortion-free benchmark. The main parameter values are:

$$\alpha = 0.283, \qquad \gamma = 0.567, \qquad \beta = 0.96, \qquad \delta = 0.08,$$

$$c_e = 1, \qquad c_f = 0, \qquad \lambda = 0.10, \qquad s \in [1, 3.98].$$

A useful benchmark relation in the undistorted economy is

$$\frac{n_i}{n_j} = \left(\frac{s_i}{s_j}\right)^{\frac{1}{1-\gamma-\alpha}}.$$

- **Interpretation.** Because plant size is monotone in productivity, the observed range of establishment employment sizes in the U.S. pins down the support of s.

- **Distributional discipline.** The distribution $h(s)$ is chosen on a 100-point log-spaced grid so that the model matches the U.S. size distribution of establishments.
- **Role of $c_f = 0$.** In the benchmark, all establishments that draw a positive s operate. This suppresses one margin of selection and keeps the paper's quantitative message centered on size distortions and factor reallocation.

3.8 How the paper measures the effect of distortions

The paper studies several policy packages. In the baseline quantitative exercises it chooses taxes and subsidies so that *aggregate capital accumulation remains unchanged*. When aggregate labor and capital are both unchanged, relative output and relative measured TFP move one-for-one.

- **Why hold aggregate capital fixed?** The authors want to isolate pure reallocation effects rather than the already well-studied effect of taxes on aggregate capital accumulation.
- **Measured** TFP**.** Aggregate TFP is output divided by a Cobb-Douglas composite of aggregate capital and labor using their income shares.
- **Central intuition.** In the undistorted economy, establishments of a given productivity type operate at their optimal scale. Distortions make subsidized plants too large and taxed plants too small. With decreasing returns, this lowers aggregate efficiency even if the total amount of factors is unchanged.

4 Identification and instruments (what makes the estimates credible)

This section needs to be interpreted somewhat differently than in a standard econometric paper. There are no external instruments here. Credibility instead comes from the calibration discipline and from how tightly the model maps plant-level distortions into aggregate outcomes.

4.1 What is disciplined directly by data

- **Factor shares.** The split between α and γ comes from capital and labor income shares.
- **Intertemporal side.** β is chosen to match a 4 percent real rate of return, and δ is chosen to match an investment-output ratio of 20 percent.
- **Heterogeneity support.** The observed employment range of U.S. establishments pins down the support of plant productivity.
- **Cross-sectional distribution.** The distribution $h(s)$ is disciplined by the U.S. establishment size distribution, not chosen freely to make the policy experiments dramatic.
- **Exit.** The annual death rate $\lambda = 0.10$ is chosen to match job destruction and exit rates in U.S. manufacturing.

4.2 What the experiments are designed to isolate

- **Pure reallocation effects.** The main experiments neutralize changes in aggregate capital accumulation, so the resulting output losses can be read as efficiency losses from plant-level misallocation.
- **Noise versus systematic distortions.** The paper contrasts uncorrelated distortions with distortions correlated with productivity. This is crucial because it shows that not every plant-specific wedge is equally destructive.
- **Cost of generating misallocation.** By reporting Y_s/Y, S/Y, and the subsidy rates needed to maintain a fixed aggregate capital stock, the paper does not just report output losses. It also shows how much fiscal or quasi-fiscal intervention is needed to create those losses.

4.3 Why decreasing returns are central

- In a model with decreasing returns, plant size is economically meaningful. There is an optimal scale conditional on s.
- Distortions do not merely reshuffle accounting measures; they move plants away from their efficient size.
- This is why even distortions unrelated to productivity can reduce TFP: they create a nondegenerate size distribution within a productivity class.

4.4 Relation to the literature

- **Relative to representative-firm growth models.** The paper adds within-country allocation across establishments to the usual focus on aggregate accumulation and aggregate prices.
- **Relative to Hopenhayn-style industry dynamics.** It uses plant heterogeneity and entry/exit, but the object of interest is not just firm dynamics; it is the macroeconomic cost of distortions.
- **Relative to size-dependent policy models.** The paper differs from Guner, Ventura, and Xu by studying a broad class of idiosyncratic distortions rather than policies directly tied to plant size thresholds.
- **Relative to later misallocation papers.** This paper is one of the foundational quantitative statements that idiosyncratic distortions can create very large aggregate TFP losses.

5 Tables and figures

5.1 Figure 1: Distribution of establishments by employment

Read:

- The benchmark calibration reproduces the cumulative U.S. establishment size distribution closely.
- This matters because the strength of decreasing returns and the support of plant productivity are what determine how responsive factor demands are to taxes and subsidies.

- The figure is the main visual validation that the benchmark economy is not arbitrary: it is built to look like the U.S. cross section of plant sizes.

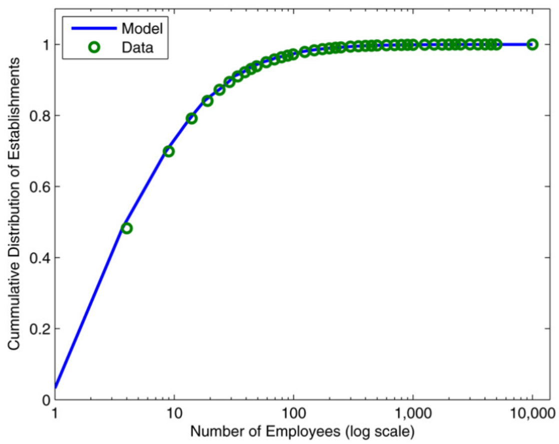

Fig. 1. Distribution of establishments by employment—model vs. data.

Table 1
Benchmark calibration to US data

Parameter	Value	Target
α	0.283	Capital income share
γ	0.567	Labor income share
β	0.96	Real rate of return
δ	0.08	Investment to output ratio
c_e	1.0	Normalization
c_f	0.0	Benchmark case
λ	0.1	Annual exit rate
s range	[1, 3.98]	Relative establishment sizes
$h(s)$	see Fig. 1	Size distribution of establishments

5.2 Table 1: Benchmark calibration to U.S. data

Read:

- The calibration is deliberately parsimonious: $\alpha = 0.283$, $\gamma = 0.567$, $\beta = 0.96$, $\delta = 0.08$, $c_e = 1$, $c_f = 0$, and $\lambda = 0.10$.
- The support of plant productivity is normalized to $[1, 3.98]$, which is chosen so the implied range of establishment sizes lines up with the U.S. data.
- The table makes clear that the striking quantitative results later in the paper are not coming from an exotic calibration.

5.3 Table 2: Distribution statistics of the benchmark economy

Read:

- The model reproduces the strong skewness in plant size: 56% of establishments have fewer than 5 workers, but they account for only 8% of output, labor, and capital.
- By contrast, plants with at least 50 workers are only 5% of establishments but account for 58% of output, labor, and capital.
- Because the technology is Cobb-Douglas with common exponents, the distributions of output, labor, and capital coincide across establishment size classes in the benchmark.

5.4 Table 3: Effects of idiosyncratic distortions — uncorrelated case

Read:

Table 2
Distribution statistics of benchmark economy

	Establishment size (number of employees)		
	< 5	5 to 49	⩾ 50
Share of establishments	0.56	0.39	0.05
Share of output	0.08	0.34	0.58
Share of labor	0.08	0.34	0.58
Share of capital	0.08	0.34	0.58
Average employment	2.4	15.5	183.0

Table 3
Effects of idiosyncratic distortions—uncorrelated case

Variable	τ_t			
	0.1	0.2	0.3	0.4
Relative Y	0.98	0.96	0.93	0.92
Relative TFP	0.98	0.96	0.93	0.92
Relative E	1.00	1.00	1.00	1.00
Y_s/Y	0.72	0.85	0.93	0.97
S/Y	0.05	0.08	0.09	0.10
τ_s	0.06	0.09	0.10	0.11

- When half the plants are taxed and half subsidized, but the distortion is unrelated to productivity, aggregate losses are moderate. At a tax rate of 40%, relative output and relative TFP fall to 0.92.
- Entry stays fixed at 1.00 because the policy package is designed to keep aggregate capital unchanged and, in this case, average establishment scale unchanged as well.
- The mechanism is purely within-productivity-class size dispersion: subsidized plants become too large and taxed plants too small, even though there is no direct shift of resources toward low-productivity types.
- The output share of subsidized establishments rises from 0.72 to 0.97 as the tax rate rises from 10% to 40%, even though the subsidy rate itself only rises from 6% to 11%.

5.5 Table 4: Relative TFP under uncorrelated distortions for different taxed shares

Read:

- The fraction of plants subject to the tax matters enormously.
- If only 50% of plants are taxed, a 40% tax lowers TFP to 0.92. But if 90% of plants are taxed and only 10% are subsidized, the same tax lowers TFP to 0.74.
- The reason is that when only a few plants are subsidized, they must receive very large effective support to keep aggregate capital fixed, so resources become much more concentrated in the wrong-sized plants.

Table 4
Relative TFP—uncorrelated distortions

Fraction of establishments taxed (%):	τ_t			
	0.1	0.2	0.3	0.4
90	0.92	0.84	0.78	0.74
80	0.95	0.89	0.84	0.81
60	0.98	0.94	0.91	0.89
50	0.98	0.96	0.93	0.92
40	0.99	0.97	0.95	0.94
20	1.00	0.99	0.98	0.97
10	1.00	0.99	0.99	0.99

Table 5
Effects of idiosyncratic distortions—correlated case

Variable	τ_t			
	0.1	0.2	0.3	0.4
Relative Y	0.90	0.80	0.73	0.69
Relative TFP	0.90	0.80	0.73	0.69
Relative E	1.00	1.00	1.00	1.00
Y_s/Y	0.42	0.67	0.83	0.92
S/Y	0.17	0.32	0.43	0.49
τ_s	0.40	0.48	0.52	0.53

5.6 Table 5: Effects of idiosyncratic distortions — correlated case

Read:

- This is the headline quantitative result of the paper. When low-productivity plants are subsidized and high-productivity plants are taxed, the efficiency losses become large.

- With half the plants taxed and a 40% tax, relative output and relative TFP fall to 0.69, i.e. a 31% decline.
- Subsidized establishments account for 92% of output in that case, and the subsidy bill reaches 49% of output.
- The key message is that systematic misallocation across productivity types is vastly more damaging than nonsystematic plant-level noise.

5.7 Table 6: Relative TFP under correlated distortions for different taxed shares

Read:

- Again, the fraction of establishments taxed matters a lot.
- If 90% of establishments are taxed at 40%, relative TFP drops to 0.51, i.e. almost a 49% loss.
- Even when only 20% of plants are taxed, the same 40% tax still lowers TFP to 0.81.
- This table is the clearest numerical illustration of the paper's opening claim that idiosyncratic distortions can produce TFP losses in the 30% to 50% range.

Table 6
Relative TFP—correlated distortions

Fraction of establishments taxed (%):	τ_t			
	0.1	0.2	0.3	0.4
90	0.81	0.66	0.56	0.51
80	0.84	0.70	0.62	0.57
60	0.88	0.77	0.69	0.65
50	0.90	0.80	0.73	0.69
40	0.92	0.82	0.76	0.72
20	0.95	0.89	0.84	0.81
10	0.97	0.92	0.88	0.86

Table 7
Taxing all but some exempt establishments ($\tau_t = 0.40$)

Variable	Establishments exempt (%)				
	10	30	50	70	90
Relative Y	0.66	0.65	0.65	0.69	0.78
Relative TFP	0.85	0.80	0.78	0.79	0.85
Relative E	0.42	0.47	0.53	0.62	0.75
Relative w	0.42	0.47	0.53	0.62	0.75
Relative K	0.42	0.47	0.53	0.62	0.75
Y_s/Y	0.10	0.31	0.52	0.73	0.89

5.8 Table 7: Taxing all but some exempt establishments (non-constant aggregate capital)

Read:

- Here the authors no longer offset the aggregate capital accumulation effect. Instead, they tax all but a subset of exempt establishments and rebate revenue lump-sum.
- Output falls more than in the fixed-capital experiments because capital accumulation, wages, and entry all fall. For example, when 50% of establishments are exempt, relative output is only 0.65, while relative TFP is 0.78.
- The paper's point is subtle: once aggregate capital is allowed to respond, output can fall more but measured TFP can fall less, because there is less reallocation toward subsidized low-productivity plants than in the benchmark correlated-distortion case.

5.9 Table 8: Idiosyncratic distortions to capital rental rates

Read:

- Distortions implemented through capital costs deliver a similar message to output wedges, but with some differences.

- In the uncorrelated case, a capital tax of 50% lowers relative TFP to 0.97; in the correlated case, a tax of 100% lowers relative TFP to 0.82.
- Relative entry now moves one-for-one with output and TFP, which did not happen in the main fixed-capital output-tax exercises.
- Because capital wedges also distort plant capital-labor ratios directly, they open an additional channel beyond the pure size distortion of the baseline output-tax case.

Table 8
Idiosyncratic distortions to capital rental rates

	Uncorrelated		Correlated	
	$\tau_t = 0.50$	$\tau_t = 1.00$	$\tau_t = 0.50$	$\tau_t = 1.00$
Relative Y	0.97	0.95	0.89	0.82
Relative TFP	0.97	0.95	0.89	0.82
Relative E	0.97	0.95	0.89	0.82
Y_s/Y	0.74	0.83	0.33	0.46
S/Y	0.03	0.04	0.10	0.14
τ_s	0.14	0.15	0.51	0.51

Table 9
Idiosyncratic distortions—output vs. wages

	Uncorrelated		Correlated	
	Output	Wages	Output	Wages
Relative Y	1.14	0.84	0.65	0.58
Relative TFP	0.98	0.89	0.66	0.67
Relative K	1.70	0.84	0.96	0.58
Relative E	1.70	0.84	0.96	0.58
Relative w	1.70	1.67	0.96	1.00
Y_s/Y	1.00	0.98	0.97	0.79
S/Y	0.50	0.56	0.48	0.45

5.10 Table 9: Output distortions versus wage distortions

Read:

- Labor taxes and subsidies cannot be tuned to keep aggregate capital exactly constant, so this table compares symmetric ±50% wage distortions with the same symmetric output distortions.
- In the uncorrelated case, wage distortions are more damaging for TFP than output distortions: relative TFP is 0.89 under wage wedges versus 0.98 under output wedges.
- In the correlated case, relative TFP is similar under both bases, about 0.66 to 0.67, but output falls more under wage distortions because aggregate capital falls more sharply.
- The broader lesson is that the exact tax base matters quantitatively, but the big-picture conclusion — idiosyncratic distortions generate large aggregate losses — is very robust.

6 Short summary

- **Conclusion.** The paper shows that the allocation of resources across heterogeneous establishments can be a quantitatively major determinant of aggregate productivity.
- **Main mechanism.** Distortions that vary across plants change their effective prices, shift factor demands, and move plants away from the efficient scale implied by their productivity.
- **Main quantitative lesson.** Distortions that are systematically correlated with productivity can reduce output and measured TFP by about one-third to one-half, even when aggregate capital accumulation is held fixed.
- **Broader implication.** Misallocation is not a residual side issue. It is a first-order channel through which policy can affect capital accumulation, relative prices, and measured TFP across countries.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that idiosyncratic policy distortions can produce large aggregate losses even in a very standard neoclassical environment.
- These losses arise without requiring aggregate price distortions or changes in the aggregate quantity of factors.
- Once heterogeneous plants and decreasing returns are introduced, who receives capital and labor becomes as important as how much capital and labor the economy has in total.

7.2 Economic intuition

- In an undistorted economy, more productive establishments are larger, but not arbitrarily larger; decreasing returns pin down an efficient size for each productivity level.
- Subsidizing some plants and taxing others creates the wrong size distribution. Subsidized plants become too large, taxed plants too small.
- If the subsidies go to low-productivity plants and the taxes fall on high-productivity plants, the distortion is especially severe because the economy not only mis-scales plants within a productivity class but also moves factors toward intrinsically less productive uses.
- This is why the paper gets much larger losses in the correlated case than in the uncorrelated case.

7.3 Takeaway

This paper is one of the cleanest calibrated demonstrations that misallocation can be a first-order source of low measured productivity. Its most important contribution is conceptual: it changes the way one reads cross-country income differences. Poor economies may not only have too little capital or weak frontier technologies; they may also allocate existing resources very badly across producers. Later work on plant-level wedges, misallocation, and productivity growth is best understood as building on this central message.